

Canonical Integration Fragment: Unequal-Twist Jets

Additive preview for ProveIt

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1 Unequal-twist jets: exact expansions and finite functional closure

Retain the twisted Fourier convention

$$\widehat{K}_\alpha^\theta(n) = (2\pi i(n + \theta))^{-\alpha}, \quad L_{n,\theta} = \log(2\pi|n + \theta|) + \frac{i\pi}{2} \operatorname{sgn}(n + \theta).$$

In this section write $J_m^{r,\theta} = (-1)^m \partial_\alpha^m K_\alpha^\theta|_{\alpha=r}$. Its multiplier is $L_{n,\theta}^m (2\pi i(n + \theta))^{-r}$. Every Fourier mode is retained. In particular $K_0^\theta = \delta_0$. The finite integer-order partial-fraction theorem does not by itself permit differentiation in those discrete order labels. A convergent identity in arbitrary orders supplies the required extension.

Theorem 1.1 (Central expansion and every mixed order jet). *Let $\theta_j \in (0, 1)$ and choose $c \in (0, 1)$ with $R = \max_j |\theta_j - c| < \eta = \min(c, 1 - c)$. The midpoint of the extreme twists always satisfies this condition. Put $d_j = \theta_j - c$, $A = \sum_j \alpha_j$, and define the polynomials h_k by*

$$\sum_{k \geq 0} h_k(\boldsymbol{\alpha}; \mathbf{d}) t^k = \prod_j (1 - d_j t)^{-\alpha_j}, \quad (1.1)$$

$$h_0 = 1, \quad kh_k = \sum_{\ell=1}^k \left(\sum_j \alpha_j d_j^\ell \right) h_{k-\ell}. \quad (1.2)$$

Then, locally normally in the distribution topology,

$$\stackrel{d}{*}_{j=1} K_{\alpha_j}^{\theta_j} = \sum_{k \geq 0} (-2\pi i)^k h_k(\boldsymbol{\alpha}; \mathbf{d}) K_{A+k}^c. \quad (1.3)$$

All order derivatives are allowed. In particular, with $M = \sum_j m_j$, $R_* = \sum_j r_j$, and the usual multi-index conventions,

$$\stackrel{d}{*}_{j=1} J_{m_j}^{r_j, \theta_j} = \sum_{k \geq 0} (-2\pi i)^k \sum_{q \leq m} (-1)^{|q|} \binom{\mathbf{m}}{\mathbf{q}} (\partial^q h_k)(\mathbf{r}; \mathbf{d}) J_{M-|q|}^{R_*+k, c}. \quad (1.4)$$

Proof. At each Fourier mode, $L_{n,\theta_j} = L_{n,c} + \log(1 + d_j/(n + c))$ because the ratio is positive. Expand the binomial factors and use $(n + c)^{-k} = (2\pi i)^k (2\pi i(n + c))^{-k}$. For $R < R_0 < \eta$, Cauchy's estimate applied to (1.1) gives $|h_k| \leq CR_0^k$ on compact order sets. The resulting modewise absolute sum has a common polynomial growth bound in n . Pairing against rapidly decreasing smooth Fourier coefficients proves normal convergence and Fourier uniqueness proves (1.3). Cauchy's formula in the order variables and Leibniz's rule give (1.4), including its sign. $\square$

All coefficients are finite Pochhammer derivatives. With unsigned Stirling numbers $\begin{bmatrix} k \\ q \end{bmatrix}$,

$$\partial_u^q \frac{(u)_k}{k!} \Big|_{u=0} = \frac{q!}{k!} \begin{bmatrix} k \\ q \end{bmatrix}.$$

For positive integer r , the same derivative at r is $(r)_k/k!$ times the complete Bell polynomial in $\mathcal{H}_1, -\mathcal{H}_2, 2!\mathcal{H}_3, \dots$, where $\mathcal{H}_j = H_{r+k-1}^{(j)} - H_{r-1}^{(j)}$. Setting $(0)_k = 0$ before differentiation would erase nonzero terms.

For arbitrary real nonintegral twists, retain a finite exceptional Fourier head exactly and apply the same expansion only at modes with $|n+c| > \max_j |\theta_j - c|$. This restores all branches and gives an entire identity without requiring a common frequency interval.

Theorem 1.2 (Exact finite closure criterion in the integer ladder). *Let a_j be distinct real shifts and $r_j, m_j \geq 0$ integers, omitting sites with $r_j = m_j = 0$. The sequence*

$$\prod_j (n + a_j)^{-r_j} \log^{m_j}(n + a_j)$$

is an eventual finite constant-coefficient linear combination of $(n+c)^{-s} \log^q(n+c)$, with arbitrary finitely many centres c , $s \in \mathbb{Z}$ and $q \geq 0$, if and only if all $m_j = 0$ or at most one site is active. Thus genuine mixed logarithmic twists cannot in general reduce to finitely many single-twist integer-order jets. Finite Fourier corrections and derivatives of δ_0 do not remove this obstruction.

Proof. Rational partial fractions and the one-site case give sufficiency. An eventual sequence identity in this class is an analytic identity: a nonzero difference has a convergent expansion $\sum_{p \geq p_0} y^{-p} P_p(\log y)$ of bounded logarithmic degree near infinity, and its leading nonzero polynomial cannot vanish at every large integer.

Continue the analytic identity around the finite singularities. Monodromy difference around $-a$ adds $2\pi i$ to $\log(y+a)$ and fixes other logarithms and all integer powers. If two sites carry logarithms, their mixed monodromy difference is nonzero on the product but zero on every single-centre generator. If only a carries a logarithm, of degree m , but a denominator is present at $b \neq a$, write the product as $R(y) \log^m(y+a)$. Applying the $(m+1)$ st monodromy difference forces higher log degrees at a on the putative right side to vanish. Applying the m th difference then identifies $R(y)$ with a Laurent polynomial in $y+a$. This is impossible because R has a pole at $-b$. These arguments also hold modulo finitely many Fourier modes, since the eventual-sequence argument is unchanged. $\square$

This is a functional statement in a specified integer-order span, not a numerical period-independence or fixed-Hurwitz-value theorem. Neither this criterion nor a numerical relation search resolves the remaining S_6/S_8 special-value problems.

1.1 A symmetric identity and its normalization

For $a = c - d$, $b = c + d$, $|d| < \min(c, 1 - c)$, set $\eta_k = H_{2k-1} - H_{k-1}$ and $E_m^c = J_m^{0,c}$. Then

$$E_1^a * E_1^b = E_2^c - \sum_{k \geq 1} \frac{(2\pi i d)^{2k}}{k} (J_1^{2k,c} + \eta_k K_{2k}^c). \quad (1.5)$$

Indeed, at a mode, with $u = d/(n+c)$ and $L = L_{n,c}$,

$$(L + \log(1-u))(L + \log(1+u)) = L^2 - \sum_{k \geq 1} \frac{u^{2k}}{k} (L + \eta_k).$$

For the canonical coefficients $Z_{1+u}^\theta = -\delta_0/u + \sum_{m \geq 0} (-1)^m U_m^\theta u^m / m!$, Gamma expansion gives $E_1^\theta = -U_0^\theta - \gamma\delta_0$ and $E_2^\theta = 2U_1^\theta + 2\gamma U_0^\theta + (\gamma^2 - \zeta(2))\delta_0$. Hence

$$U_0^a * U_0^b = 2U_1^c + 2\gamma U_0^c - \gamma(U_0^a + U_0^b) - \zeta(2)\delta_0 - \sum_{k \geq 1} \frac{(2\pi i d)^{2k}}{k} (J_1^{2k,c} + \eta_k K_{2k}^c). \quad (1.6)$$

The point mass is necessary, including when $a = b = c$.

1.2 Ordinary Gamma integrals and scalar finite parts

Write $P_\theta = J_1^{1,\theta}$, $S_0 = 0$, and $S_k = \sum_{\ell=1}^k \eta_\ell / \ell$. Dividing the preceding mode identity by $\lambda_{n,c}^2(1-u^2)$ yields

$$\int_0^1 P_a(t) P_b(\{x-t\}) dt = \sum_{k \geq 0} (2\pi i d)^{2k} (J_2^{2k+2,c}(x) - H_k J_1^{2k+2,c}(x) - S_k K_{2k+2}^c(x)). \quad (1.7)$$

This is an ordinary integral: $P_\theta \in L^2$ since its coefficients are $L_{n,\theta}/\lambda_{n,\theta}$. The coefficient double sum on the right is absolutely convergent, dominated after summing over k by $O((1 + \log^2(2 + |n|))/(1 + |n|)^2)$, so equality is continuous and holds at every circular argument. For $\theta = p/q$, with $z = e^{-2\pi i p/q}$, its factors are explicitly

$$P_{p/q}(x) = e^{-2\pi i p x/q} \left\{ \sum_{\ell=0}^{q-1} z^\ell \log \Gamma\left(\frac{x+\ell}{q}\right) - \frac{\gamma + \log q}{1-z} \right\}, \quad 0 < x < 1. \quad (1.8)$$

To derive this representative, use $\Phi(z, s, x) = q^{-s} \sum_{\ell} z^\ell \zeta(s, (x+\ell)/q)$, $\zeta'(0, y) = \log \Gamma(y) - \frac{1}{2} \log(2\pi)$, and the derivative of $1/\Gamma(\alpha)$ at $\alpha = 1$.

For $a, b > 0$ define the sharp-cutoff constants

$$C_{m,n}(a, b) = \lim_{N \rightarrow \infty} \left\{ \sum_{j=0}^{N-1} \frac{\log^m(j+a) \log^n(j+b)}{j+a} - \frac{\log^{m+n+1} N}{m+n+1} \right\}.$$

Subtracting the diagonal summand leaves an absolutely convergent series. A unilateral binomial expansion and the Pochhammer jet at zero therefore give, with $M = m+n$ and $|b-a| < a$,

$$C_{m,n}(a, b) = \gamma_M(a) + (-1)^M \sum_{k \geq 1} (a-b)^k \sum_{q=1}^{\min(n,k)} \binom{n}{q} \frac{q!}{k!} \left[\frac{k}{q} \right] \zeta^{(M-q)}(k+1, a). \quad (1.9)$$

All zeta superscripts here are order derivatives. For other positive shifts, split off a finite head and translate both shifts by the same integer until the displayed condition holds. Differentiating the convergent difference sum and applying partial fractions gives

$$\partial_b C_{m,n}(a, b) = \frac{n}{b-a} (C_{m,n-1}(a, b) - C_{n-1,m}(b, a)), \quad n \geq 1, \quad b \neq a, \quad (1.10)$$

$$C_{0,1}(a, b) = \gamma_1(a) + \int_a^b \frac{\psi(t) - \psi(a)}{t-a} dt, \quad (1.11)$$

$$\int_a^b C_{0,1}(a, t) dt = (b-a) C_{0,1}(a, b) - \log \Gamma(b) + \log \Gamma(a) + (b-a) \psi(a). \quad (1.12)$$

The collision singularities in these divided differences are removable. The last equality is integration by parts.

1.3 The unilateral family and transverse ray constants

For positive a_j , choose $c > 0$ with $|a_j - c| < c$. The same coefficient polynomials, now with $d_j = a_j - c$, give

$$\mathcal{Z}(\boldsymbol{\alpha}; \mathbf{a}) := \sum_{n \geq 0} \prod_j (n + a_j)^{-\alpha_j} = \sum_{k \geq 0} (-1)^k h_k(\boldsymbol{\alpha}; \mathbf{d}) \zeta(A + k, c). \quad (1.13)$$

The left series initially requires $\operatorname{Re} A > 1$; the right is its meromorphic continuation. The only possible hyperplanes are $A = 1 - k$, and their residues are $(-1)^k h_k|_{A=1-k}$. Normal convergence of the large- k tail follows from $|h_k| \leq C R_0^k$ for $\max |d_j| < R_0 < c$ and $|\zeta(A + k, c)| \leq C c^{-k}$ on compact order sets.

For complex weights w_j , put $W = \sum_j w_j \neq 0$ and $\mathcal{Z}_{\mathbf{w}}(t) = \mathcal{Z}(t\mathbf{w}; \mathbf{a})$. Then the transverse ray has a removable singularity at zero and

$$\mathcal{Z}_{\mathbf{w}}(0) = \frac{1}{2} - \frac{\sum_j w_j a_j}{W}, \quad (1.14)$$

$$\mathcal{Z}'_{\mathbf{w}}(0) = \sum_j w_j \log \Gamma(a_j) - \frac{W}{2} \log(2\pi). \quad (1.15)$$

To prove this, isolate $k = 0, 1$ in (1.13). With $D = \sum_j w_j d_j$, these terms are $\zeta(Wt, c) - tD\zeta(1 + Wt, c)$; the second has constant $-D/W$ and linear coefficient $-D\gamma_0(c)$. For $k \geq 2$, $h_k(0) = 0$ and $\partial_t h_k(t\mathbf{w}; \mathbf{d})|_0 = \sum_j w_j d_j^k / k$. Summing the convergent Taylor series

$$\log \Gamma(c + d) - \log \Gamma(c) - d\psi(c) = \sum_{k \geq 2} \frac{(-1)^k d^k}{k} \zeta(k, c)$$

gives (1.15), using $\gamma_0(c) = -\psi(c)$. The ray value generally depends on direction: for shifts 1, 2, the rays with weights (1, 1) and (2, 1) give -1 and $-5/6$. Thus a joint value is not being assigned at the multivariable origin.

The full continuation report also proves the unilateral multishift Lerch expansion, all ordinary primitives, the meromorphic polar hyperplanes, every transverse spectral-ray jet, and the pole-aware cyclotomic Stieltjes filter. Its proofs and execution records retain separate status: 824 finite exact assertions and 96 numerical comparisons at each of two working precisions corroborate, but do not replace, the analytic arguments. No global priority or proof-assistant claim is made.